RESILIENT FUTURES LAB

02

Risk numbers and misleading claims

Read the evidence behind a claim

Percentages	**Chance and loss**
Graphs and samples	**Claims and evidence**

A learning booklet Resilient Futures Lab

Choose the route that fits your experience.

Work through the examples, discuss your reasoning, and complete the final case. A calculator, paper and a pencil are enough.

English edition October 2026

EUonAIR project

01 THE WHOLE GROUP

A percentage needs a denominator

A message says that "95% of deliveries arrive on time". To interpret it, ask how many deliveries were counted, during which period, and whose deliveries were included. The **denominator** is the total number used in a fraction.

$$\text{Percentage} = \frac{\text{number of relevant cases}}{\text{total number of cases}} \times 100.$$

5 late deliveries
among 100 deliveries

$5/100 = 5\%$ late
$95/100 = 95\%$ on time

Classroom example. These dots describe an invented delivery record. Each dot represents one delivery. The same percentage could arise from 1 late delivery in 20, and a small record gives less information about the service over time.

A useful claim identifies the group, the period and the source. A seller may show selected favourable weeks, so the method of choosing the data also matters.

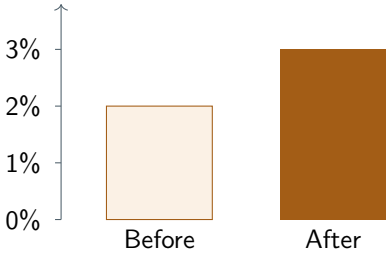
Try it A record shows 15 late deliveries out of 300. Calculate the percentage arriving late and the percentage arriving on time.

02 CHANGES IN PERCENTAGES

Percentages and percentage points

A percentage change compares a difference with the starting value. A change in **percentage points** subtracts one percentage from another.

Worked example. An illustrative service fee rises from 2% to 3% of each payment. The increase is $3 - 2 = 1$ percentage point. Relative to the starting fee, the increase is $(3 - 2)/2 = 50\%$.



On a EUR 100 payment
Before: EUR 2
After: EUR 3
Extra cost: EUR 1

These descriptions answer different questions. Percentage points describe the gap between rates. The relative change describes how large that gap is compared with the original rate.

The starting value matters in ordinary prices too: a transport pass rising from EUR 20 to EUR 24 increases by $4/20 = 20\%$.

Try it A scholarship application success rate rises from 40% to 50%. Find the change in percentage points and the relative percentage increase.

03 PRICE CLAIMS

Follow the reference price

Successive discounts apply to the price remaining after each step. For an illustrative EUR 100 jacket, a 20% reduction followed by a further 10% reduction gives:



The total saving is EUR 28, or 28% of the original price. Delivery charges and compulsory fees also belong in the final comparison.

A real European case

In September 2024, the Court of Justice of the EU considered Aldi Süd's advertised reductions. Its press release illustrated bananas priced at EUR 1.29, compared with a crossed-out EUR 1.69, and the lowest price in the preceding 30 days was already EUR 1.29. The Court explained that the announced reduction must use the relevant lowest prior price. [R2]

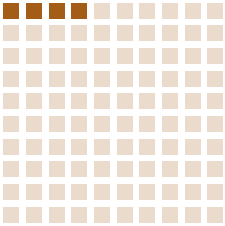
This case shows why a striking discount needs a meaningful baseline. For everyday comparisons, record the product, quantity, earlier price and complete payable price.

Try it An illustrative backpack falls from EUR 80 by 25%, then by a further 10%. Delivery costs EUR 4. What is the final payable price?

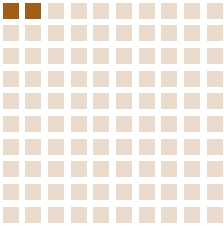
04 DESCRIBING RISK

Show the absolute numbers

An illustrative account-safety trial records unauthorised-access incidents during one month. Eight of 200 accounts using the old procedure experience at least one incident, compared with 4 of 200 using the new procedure.



Old: 4 per 100



New: 2 per 100

The observed rates are 4% and 2%. The **absolute decrease** is 2 percentage points. The **relative decrease** is $(4-2)/4 = 50\%$. Over equal groups of 200, the difference is four affected accounts.

A headline saying "risk halved" describes the relative decrease. The original rate, time period and number of observations help readers judge its practical size. Small studies can produce unstable estimates, and the way accounts entered the groups affects interpretation.

Try it A late-payment rate falls from 6% to 3%. Give the absolute and relative decreases. How many fewer late payments does this represent per 1,000 comparable payments?

Further practice | ages 18–24

Explain why this observed trial alone gives limited information about the new procedure's causal effect. Consider group selection and other security changes.

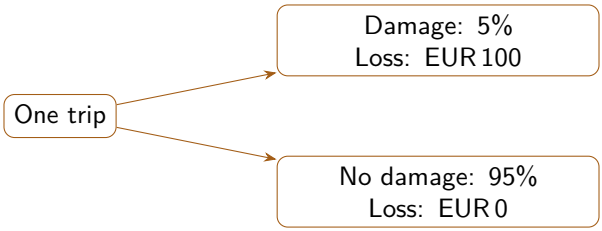
05 UNCERTAIN COSTS

Expected loss and a cash buffer

A probability describes how likely an outcome is under specified conditions. The **expected loss** combines the possible losses with their probabilities. For one possible loss:

$$\text{Expected loss} = \text{probability} \times \text{loss amount}.$$

Classroom example. Suppose a borrowed device has an assumed 5% chance of accidental damage during a trip, costing EUR 100. The expected loss is $0.05 \times 100 = \text{EUR}\,5$.



Across many comparable trips, the average loss could approach EUR 5 if the assumed model holds. For one trip, the model allows a loss of EUR 100 or EUR 0. A EUR 5 reserve would leave a EUR 95 gap after damage.

Compare both the average cost and the amount that must be available if the adverse event occurs. Real probabilities require credible evidence.

Try it Using an assumed 2% chance of a EUR 300 loss, calculate the expected loss. What funding gap remains after that loss if the reserve is EUR 40?

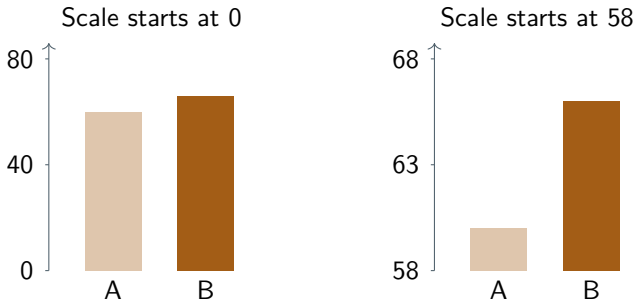
Further practice | ages 18–24

A protection plan costs EUR 9 and fully covers that loss. Compare expected costs, immediate affordability and the terms that require checking.

06 GRAPHS

Read the scale before the shape

Two charts can display the same observations and create different first impressions. The illustrative weekly travel costs below rise from EUR 60 to EUR 66: an increase of EUR 6, or 10%.



The shortened scale makes the difference occupy much more of the chart. Bar length represents magnitude most clearly when the baseline is zero. A focused scale can be useful in a line chart when its limits are clearly stated.

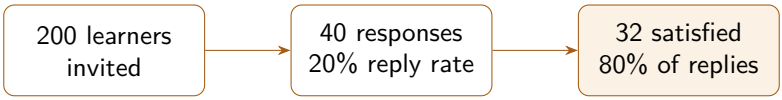
Check the unit, axis limits, intervals and dates. A chart covering two selected weeks may conceal seasonal changes or a longer trend. Read the data values alongside the picture.

Try it An advertisement plots a score of 80 and then 84 on an axis running from 79 to 85. Calculate the relative increase and describe how you would redraw the bars.

07 EVIDENCE QUALITY

Who answered and what changed

An invented training provider invites all 200 past learners to answer a satisfaction survey. Forty reply and 32 of those report being satisfied.



The 80% describes respondents. The views of the other 160 learners remain unknown. Enthusiastic learners, unhappy learners or people with spare time may be more likely to answer. This is **selection bias** when participation systematically changes the picture.

Causation means that a change in one factor produces a change in another. If course participants later earn more, their previous skills, motivation or local job opportunities may explain some of the difference. A fair comparison considers these factors and measures outcomes consistently.

Try it A provider shows five successful former learners in a video. Name two pieces of information needed to judge the typical learner's outcome.

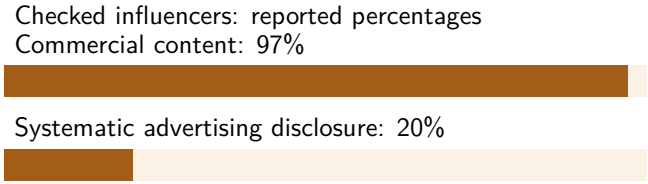
Further practice | ages 18–24

Propose a fair evaluation comparing a course group with another group. State one outcome, a follow-up period and a way to address initial differences.

08 CLAIMS AND INCENTIVES

Ask what supports the promise

In results published in February 2024, European consumer authorities reported checks of 576 influencers. Some 97% posted commercial content, while around 20% systematically identified it as advertising. These figures describe the checked group. [R3]



A payment, free product or sales commission may influence what a creator recommends. Look for the commercial relationship, the original evidence and the assumptions behind a claim. ESMA's 2024 guidance also stresses reliable sources and disclosure of conflicts in investment recommendations. [R4]

Classroom example. A promotional message promises a 10% increase every month. Repeating the claim for a year implies $1.10^{12} \approx 3.14$: more than three times the starting amount. This calculation tests the size of the promise; credible evidence must still support it.

Try it Write two questions you would ask before trusting a creator's claim that a paid course "doubles your chance of work".

Further practice | ages 18–24

Calculate the implied annual percentage increase from 10% growth every month. List the missing assumptions about costs, losses and verification.

SOURCES

References and further reading

The mathematical examples and final case use invented data for learning. The Aldi Süd case and influencer checks are documented events. The learning aims draw on the European financial competence framework, especially its treatment of risk and informed decisions. [R1]

[R1] European Union and OECD (2023). Financial competence framework for children and youth in the European Union. See Risk and reward and Financial landscape.

[R2] Court of Justice of the European Union (2024). Judgment in Case C-330/23, Aldi Süd. Press release 152/24, 26 September.

[R3] European Commission (2024). Sweep on influencers. Findings from checks of 576 influencers, published in February 2024.

[R4] European Securities and Markets Authority (2024). Warning for people posting investment recommendations on social media, 6 February.

Sources checked 30 October 2026. Each linked title opens an original institutional source. Printed copies can be used entirely offline.

FINAL CASE

A course advertisement for students

Fictional case. Kateryna is comparing short online skills courses while preparing applications for part-time work. A creator promotes CareerStep, a fictional provider, and receives a payment for each enrolment. The post says: "25% off and 80% learner satisfaction. Improve your chances of work."

The provider supplies the following information:

Item	Figure
Advertised base course price	EUR 120
Discount from base course price	25%
Compulsory registration fee	EUR 10
Lowest earlier course price in 30 days	EUR 100
Former learners invited to a survey	200
Survey responses	40
Satisfied respondents	32
Employment rate in an earlier cohort	40%
Employment rate in a later cohort	50%

The earlier course price excludes the same EUR 10 registration fee. Both employment rates refer to six months after completion. The cohorts studied in different years; the provider supplies no comparison group or starting characteristics. A bar chart of the employment rates begins at 39%.

Kateryna has set a EUR 105 limit for the complete purchase. She wants a useful course and a credible account of its results. Work only with the fictional figures on this page; write your calculations and reasoning on the following page or on separate paper.

FINAL CASE

Your calculations and judgement

1. Calculate the discounted course price and the complete payable amount. Does the purchase fit Kateryna's EUR 105 limit?
2. Compare the current complete price with the earlier complete price. Find the saving in euros and as a percentage.
3. Calculate the survey response rate and the satisfaction rate among respondents. Explain whose views remain unknown.
4. Express the employment-rate change in percentage points and as a relative percentage increase.
5. Explain how the 39% chart baseline affects the visual impression. Describe a clearer bar chart.
6. Give two reasons why the cohort figures alone provide limited evidence that the course caused the improvement. State one useful additional source of evidence.
7. Explain how the creator's payment affects your assessment. Write a three-sentence recommendation for Kateryna, identifying what she should verify before deciding.

Further practice | ages 18–24

The provider now claims 10% growth in employment chances each month for a year, starting from 40%. Calculate the implied rate and explain the mathematical problem with continuing this rule.